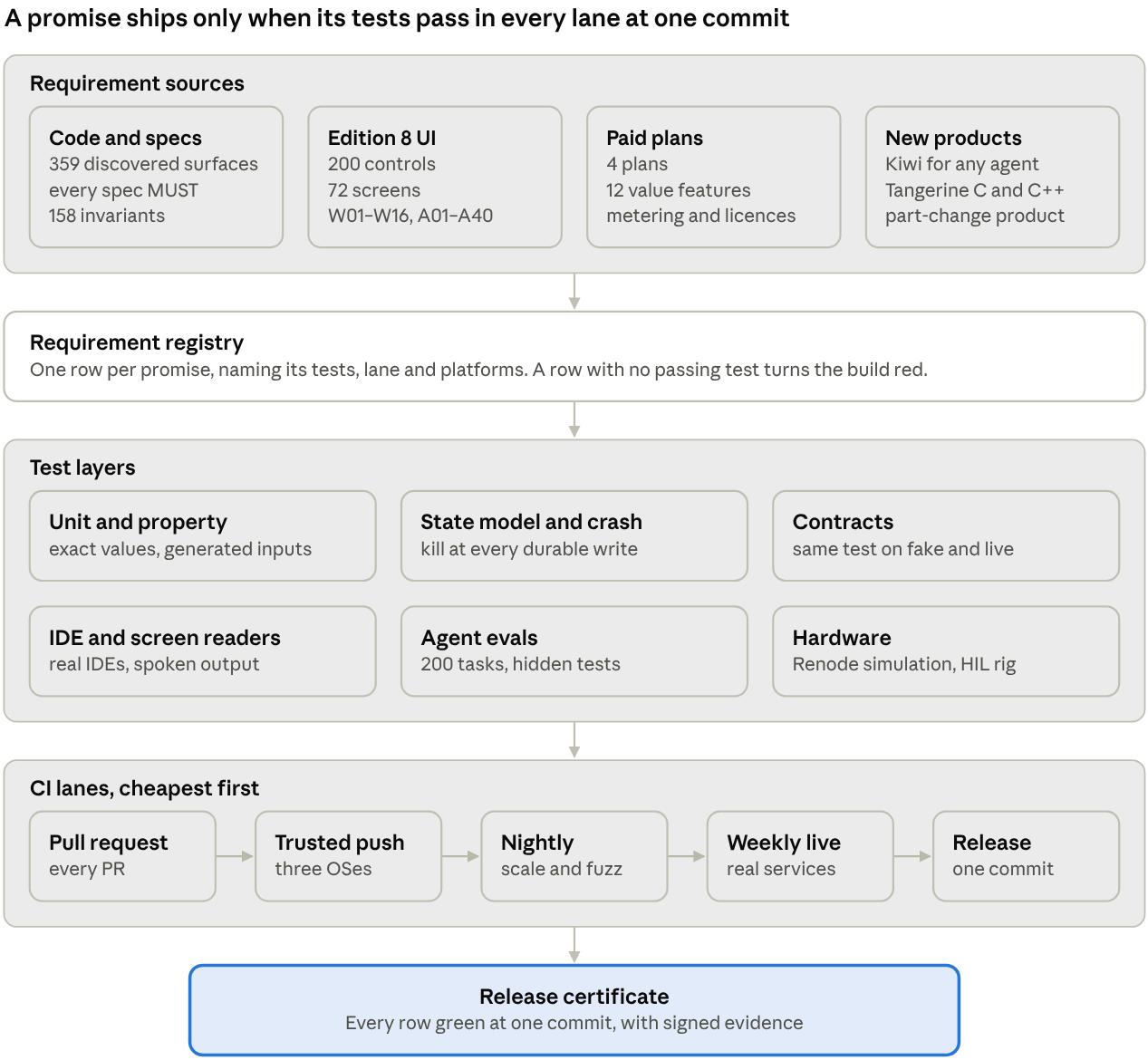
Faktor automated test programme

Oct 18, 2026 · @Aaron

This report combines a review of Faktor's latest HEAD (commit `ec50290`, 18 Oct 2026) with a test programme that gives every feature its own automated proof. The repo itself discovers 359 product surfaces; on top of those sit 200 controls, 72 screens, 16 workflows, 40 acceptance scenarios and 158 invariants.

The headline: no surface yet has its own behaviour proof in the coverage registry. All 346 registry rows delegate to five generic invariants, 17 routes added in Revision 8 have no row at all, and the HEAD fixes expose a few new defects that need tests before release.

At a glance



test programme · sources, registry, 6 layers, 5 lanes

Every promise gets a registry row, its tests run in up to five lanes, and only a full set of green results at one commit makes a release.

What changed at HEAD

Commit `ec50290` ("Fixes", 18 Oct 2026) changes 84 files: about 5,000 lines of code and tests, plus 48,000 lines of visual baselines. Most fixes come from a new 1,000-turn scale probe that drove the real daemon and found races, quadratic reads and wrong counts.

The fixes are real, but most leave a gap that still needs a test, and one introduces a new defect.

Act now:

- Remove the two committed Chromium core dumps (`qemu_chrome_20261018-133409_41.core` and `_43.core`, 89 MB each) from history. A pattern scan found no credentials in them, but every clone of the public repo downloads them.
- Replace `kind LIKE 'memory_v2:%'` with a range predicate in both memory pagers (row 5).
- Prove or close the remaining settlement race on the paths that pass no task id (row 1).
- Run the new scale probe in a scheduled lane. Nothing in CI runs it today.

Change	What it fixed	What still needs proof	Test to add
1. Settlement bound to the admitted task	A prompt starting during the previous turn's settlement folded the old goal, spend and completion gate into the new task, and could close it before its first model call	Paths that pass no task id (queued prompt, recovery, shadow watcher) look up a run row under `tx-session-<id>`. Only a policy row is written there, so they fall back to the current task. 18 other lazy `task_id()` reads remain in agent and orchestrator code	Deterministic interleaving: start prompt N+1 at every await point of turn N's settlement; task N+1 must stay byte-identical. Static scan banning lazy `task_id()` reads in settlement modules
2. Tool activation by point lookup	A `source_market` activation silently switched off once 3,200 newer internal facts existed	The `BoundExhausted` state is now dead in production code	Activation survives 1,000,000 churn rows; a corrupt value reads as failed, never as inactive

Change	What it fixed	What still needs proof	Test to add
3. Child control rows filtered by kind	Long sessions failed turns with "refusing a partial control read" past 5,000 facts	A hostile row space still stops at 25 pages	1,000,000 unrelated facts plus 5,001 control rows: typed refusal, never a truncated queue
4. Incremental ledger verification	Every session open re-decoded the whole ledger, quadratic over a turn loop	Corruption inside the already-verified prefix is caught only after a restart	Corrupt a verified row; a scheduled full sweep must fail loudly
5. Memory paging filtered in SQL	Context preparation paged through every internal fact on each turn	**New defect.** `LIKE` is case-insensitive, treats `_` as a wildcard and cannot use the index. On the same schema: 49 ms per page at 200,000 churn rows, against 0.02 ms with `kind >= 'memory_v2:' AND kind < 'memory_v2;'`	Query-plan gate naming the `(session_id, kind, key)` index; 200,000-row timing test; kinds `MEMORY_V2:x` and `memoryXv2:y` must not match
6. Workspace cache capped at 64 roots; shadow roots opened without watchers	The probe accumulated over 500 file-watcher threads	Eviction drops an arbitrary entry, possibly the hot durable root. The new test forces a 1 ms deadline, so it never runs a real watcher	1,000 shadowed turns per OS with real watchers: thread and handle counts stay flat; an in-use durable root is never evicted
7. Routing reads a 64-call tail	Each turn re-read every provider call in the session	None found	On recorded sessions, the 64-call tail gives the same routing decision as full history

Change	What it fixed	What still needs proof	Test to add
8. History paging (`beforeMs`, `beforeId`, `limit`) and a new index	History with 2,000 conversations loaded the whole table	The cursor sorts on `updated_ms`, which changes. A conversation updated mid-walk jumps to page one and later pages skip it	Update conversations during a page walk; define the rule (snapshot or re-surface) and assert it
9. Transcript search reads answer parts	Model answers were invisible to Find	Search matches raw JSON, so `text`, `parts` or a quote character match every row. `lower()` folds ASCII only, so "É" never finds "é"	Results must equal a reference search over the visible decoded text; Unicode case-folding corpus
10. Proof totals cover every row	A 10,000-criterion record reported page-local counts	None found	Totals at 0, 1, 499, 500, 501 and 10,000 rows equal the durable counts
11. Tiny-history compaction	Compacting a one-token history crashed the daemon (`clamp(1, 0)` panic)	Other routes may hide the same panic class	Fuzz every route's request body with boundary values under a panic hook that fails the run
12. Budget caps accepted on plain prompts	Caps were refused on this path after a 1-micro cap once let a paid call through	No new test proves a cap of "1" stops the first paid call here. A cap of 0 means no cap	`max_cost_micro: "1"` and `max_tokens: 1` give zero provider calls and a typed refusal; the zero-cap rule is shown and tested

Change	What it fixed	What still needs proof	Test to add
13. Task and run state tags	A finished turn showed "running" forever; verifying, blocked and waiting showed as done	The two IDEs render the tags separately	Every task state × session state pair maps to one tag, rendered the same in both IDEs
14. VS Code tool outcome merge	A denied call showed "read_file done" beside an " (unnamed tool) exit 1" bubble	Merging runs after the transcript is cut to its newest window, so a call just outside it leaves an orphan bubble. A denial with text but no exit code still shows "done"	Every tool × outcome (allowed, denied, cancelled, timed out, unknown effect), with and without exit code, never shows a pass tone; window-edge case
15. VS Code webview handshake and backpressure	Panels stayed empty when the ready message or patches were lost	Only the fixed ordering is covered	Message before listener, hide and reveal mid-refresh, reload, 100 rapid reveals: each ends showing the complete served state
16. JetBrains scrollbar button names	The accessibility audit failed on unnamed scrollbar buttons	The fix targets the Metal look-and-feel of the headless harness, not the real IDE's	Run the audit inside a real IDE with its own look-and-feel
17. Migration tests use fixed versions	Tests written as `SCHEMA_HEAD - N` silently skipped a migration when v39 landed	Other arithmetic offsets may remain	For every version from 0 to head: migrate, then compare the whole schema to a fresh head database

Change	What it fixed	What still needs proof	Test to add
18. Skills listing reports totals	Workspaces with over 200 skills silently showed 200	There is no cursor: skill 501 onward is unreachable	All 600 discovered skills are reachable
19. Two core dumps added	Not a fix	`.gitignore` covers `core.*` but not `*.core`; the hygiene scan checks a fixed list of extensions	Sniff file content (ELF core, minidump, Mach-O) and cap the size of any blob not on an allowlist
20. Another file-size exemption (`sessions.rs`, "split deferred")	Not a fix	The exemption list only grows	Ratchet: exemptions may only shrink

The additions worth keeping as gates are the scale probe, the fake Ollama server and the JetBrains EDT harness. The pull-request lane still writes a visual baseline when a platform has none, so a run can compare against a baseline it wrote itself.

Where coverage stands

Faktor's machinery tests are deep, but its per-feature proof is thin. The repo's own coverage registry marks every surface as covered by delegating it to one of five generic invariants, so it cannot show which features lack a behaviour test.

Surface	Count	Own behaviour row in the registry	Named by at least one test	What the numbers hide
Native routes	246	0. 229 delegate to "every route refuses an unauthenticated call"; 17 Revision 8 routes have no row	211	35 routes are named by no test, including credential store, rotate and delete

Surface	Count	Own behaviour row in the registry	Named by at least one test	What the numbers hide
CLI commands	43	0. All delegate to the `doctor` exit-code test	24	No test runs 8 `updater`, 10 `enterprise` and 1 `sandbox` subcommands by name
IDE contributions	56 (55 VS Code, 1 JetBrains)	0. All delegate to the client route-name drift check	Most, through the VS Code self-test	Command behaviour is not asserted, only that names match the daemon
Built-in tools	8	0. Seven delegate to the VS Code route-name check, one to a search lock invariant	All, inside crate tests	The delegates do not exercise the tools
Provider kinds	6	0. All delegate to the HTTP error classification table	All, with recorded fixtures	Live behaviour is checked only by the keyed release run
Controls	200	Exercise passes 200 of 200 in each IDE	All	The repo's own audit says 40 complete, 148 partial, 12 missing; 175 carry a contract gap. That audit is hand-written and already stale: its summary says 25, 160 and 15, and two "missing" controls now have routes

Surface	Count	Own behaviour row in the registry	Named by at least one test	What the numbers hide
Screens	72	Visual contract gate in both IDEs	All	Windows JetBrains visual records are missing; Linux baselines were written by the pipeline
Workflows	16	5 cells each, per IDE	All	The screen-reader cell checks names and roles; real NVDA, VoiceOver and Orca runs are recorded as not certified
Acceptance scenarios	40	40 of 40 in each IDE, with forbidden-state scans	All	None found
Invariants	158	Each has a unit test, a wiring test and a planted mutation	All	This is the strongest layer, yet five invariants carry all 346 delegations
Crates	51	6,880 Rust tests in total	—	`faktor-foundation` (settings, providers, vault, instructions, presets, policy) has 13,400 lines and 62 tests, the thinnest ratio in the workspace

The counts come from running the repo's own `discover-capabilities.mjs` and reading `tests/invariant-coverage.json` at HEAD. "Named by a test" means a route path, handler or command appears in test code, so it overstates real coverage.

Every surface is listed with its tests in the other tabs: `Routes (250)`, `Controls (200)`, `Screens, workflows, acceptance` and `CLI, IDE, tools, config`.

Rules every test follows

Faktor already tests its machinery hard: 6,880 Rust tests, 158 release-critical invariants each with a planted-mutation check, crash campaigns, fuzz targets and the Edition 8 IDE gates. These rules extend that discipline from invariants to every feature.

1. **One requirement registry.** Every promise the product makes gets a row: README features, each MUST in the specs, every native route and CLI command, the 200 controls, 72 screens, W01–W16, A01–A40, each paid entitlement and each new product.
   - A row names its tests, lane and platforms.
   - A gate fails the build when any row lacks a passing test at the exact commit.
   - It extends `tests/invariants.toml`, `check-invariants.mjs` and `dogfood.mjs` rather than replacing them.
2. **Every test has an oracle that can say no.** Allowed oracles: an exact expected value, a reference implementation, a known relation between two runs (metamorphic), a property over generated inputs, a reference state model, or a recorded real-world result. "It didn't crash" is not an oracle.
3. **Every claim is attacked.** Each requirement needs at least one refusal or boundary case besides the happy path. Release-critical rows also need a planted mutation the suite must catch, as invariants already do.
4. **Offline lanes are deterministic.** Injected clock, seeded randomness, scripted model responses, no network. A failure prints its seed and replays exactly.
5. **A flaky test is a bug.** It is quarantined with an owner, and no release ships while any test is quarantined.
6. **Live lanes are separate and honest.** Real models, vendors, GitHub and IDEs run in scheduled lanes with their own signed evidence. A live result is never copied to another platform or commit, matching today's certificate rules.
7. **Strength is measured.** Mutation score per crate (mutants killed ÷ mutants planted) is tracked beside coverage. I propose a floor of 80% on authority-bearing crates.

The HEAD review adds eight more rules:

8. **No delegated coverage.** A surface counts as covered only by a test of its own behaviour. Delegating to a shared invariant proves that shared property (authentication, name drift) and nothing else.
9. **Status is derived, never typed.** Control status, coverage counts and baselines are generated from code and test results on every run, as the capability manifest already is.
10. **Every fix ships two tests.** One for the exact regression and one for the whole class of the bug: all lazy task-id reads, all unbounded scans, all panicking clamps.
11. **Scale is a gate.** The scale probe runs on a schedule with curve-ratio limits; a quadratic latency curve fails even when absolute times look fine.
12. **Every new query has a plan test.** `EXPLAIN QUERY PLAN` must name the intended index; a full scan or temporary sort on a growing table fails.
13. **Exemptions only shrink.** File-size ceilings, ignored tests, quarantines and allowlists are ratchets.
14. **Baselines are reviewed artifacts.** A run never compares against a baseline it wrote; a baseline change carries the rendered differences and a reviewer.
15. **No route can panic.** Every request body is fuzzed with boundary values under a panic hook that fails the run.

Test templates by surface class

Every surface inherits the mandatory cases of its class; the tabs then add what is specific to each one. A generator reads the discovered surface list and creates one test stub per surface × case, so a new route or command arrives with failing stubs until someone writes its tests.

Class	Instances	Mandatory cases for every instance
Native route	246	Strict body: unknown, missing and mistyped fields each give a typed 400. Response matches the frozen DTO; money is a decimal string at every depth; an absent cap is `null`, never 0. Auth: missing, malformed and foreign-session credentials refused. Bounds: body size, list length, string length, negative or overflowing ids, unknown ids. Mutations: same submission id replays the same receipt; a refused call changes no durable row; two concurrent identical calls give one effect. Crash at each durable write inside the handler leaves the before or after state. Errors are typed, never a panic, never leak paths or secrets. A disabled section answers its typed 409. Both IDE clients parse the response
Event stream	2 (`/native/events` , per-session events)	Resume from every event id gives each later frame exactly once; per-session order; no cross-session frames; frame size bounds; slow consumer and reconnect storm stay bounded

Class	Instances	Mandatory cases for every instance
CLI command	43	Golden help text; exit codes for success, usage error, refusal and internal error; unknown flags fail; unknown config keys fail at start; rerun is idempotent; no network unless the command needs it; Ctrl-C mid-run leaves a consistent store
Built-in tool	8	Bad arguments refused before any effect; permission and sandbox checked before the effect; bounded output; tool output treated as data, never instructions; after a crash an unknown effect is never re-run; cancellation; lazy tools cost zero tokens when inactive
Provider adapter	6	Recorded streams with split frames, split tool-call arguments, usage, reasoning and every error status; same conversation normalizes to the same events across families; deadline, retry and Retry-After; cancel mid-stream; refuse unsupported capabilities before sending; weekly live contract run

Class	Instances	Mandatory cases for every instance
Configuration	22 daemon sections, 17 VS Code settings	Default value; each invalid value gives a typed error; unknown key fails at start; precedence and provenance across machine, user, workspace and session; applies at its declared time; a disabled feature behaves byte-identically to an absent one; secrets never in plain config; export then import round-trips
Durable table	44 in the session store, 95 in the cloud, foundation, commerce, worker, SCM and updater stores	Migration from every earlier version; crash during migration restores from its restore point; corruption fails loudly; every query has a plan test; retention and deletion cascade with an audit tombstone
Background job or watcher	Every spawned task (watchers, report schedules, GC, retention, deletion jobs, worker heartbeats)	Owned by a registry, never orphaned; bounded retries and deadline; restart resumes exactly once; shutdown leaves zero live tasks; clock jumps forward and back
IDE command, setting or view	56	Invoked from palette and keybinding; enablement conditions; effect checked against the daemon, not just the name; setting applies at its declared time; hide, reveal and reload end in the complete state

Class	Instances	Mandatory cases for every instance
Control	200	Default, scope and apply timing from its contract; empty, loading, error, disabled and stale states each rendered and recoverable; invalid input preserved; keyboard and screen reader; same semantics in both IDEs; 200% zoom and 320 px
Screen	72	Visual contract per IDE, theme and width; primary action present; focus order; no fabricated values in empty or stale states
Workflow	16	Pointer, keyboard, screen reader, compact and recovery cells per IDE, plus a daemon crash and a network loss mid-flow
Acceptance scenario	40	Printed requirement, field values and forbidden-state scan, plus one adversarial variant each
Entitlement	12 features × 6 plan cases	Enforced at its own runtime boundary; the community path stays ungated
External adapter	GitHub App, Kiwi, Tangerine, 5 marketplaces, MCP servers, SSO, update feed	The same contract test runs against the fake and the live service; hostile responses; timeouts; credentials never cross scopes
Invariant	158	Unit, wiring and planted mutation, as today; a fault campaign where a crash can break it

Shared harness

Every suite below draws on the same harness pieces. Half exist today; the rest are the first things to build.

Harness piece	Today	Add
Surface stub generator	Discovery script lists 359 surfaces	One failing test stub per surface × template case, regenerated on every run; a stub that is still failing blocks release
Scripted model responses	`FakeProvider` in crate tests; a fake Ollama server (new at HEAD) for daemon-level runs	Record and replay of real provider streams with secrets scrubbed, so any live failure becomes an offline test
Transport chaos	Recorded stream fixtures per provider family	A chaos proxy: latency, dropped connections, truncated or malformed SSE, split UTF-8, 429 and 5xx with Retry-After
Crash injection	Fault campaigns for journal, edits, CAS, scheduler, recovery and abort	A failpoint at every durable write (fsync, rename, commit); each scenario reruns, killing the process at each point in turn
Store faults	None	`SQLITE_BUSY`, disk full, read-only file system and a corrupted page injected under every store operation
Interleaving control	None	A test executor that pauses tasks at named await points and runs every ordering of two drives (the settlement race class)

Harness piece	Today	Add
Property testing	Hand-written property crates (router, economy)	`proptest` with shrinking and stored regression seeds
Concurrency checking	None	`loom` or `shuttle` model checks of the single-writer store, writer queues, admission leases and scheduler budgets
State-model testing	`accounting-modelcheck`	A reference model of the session state machine; random command sequences must leave the same state after replay
Query plans	Plan tests in 7 crates	A plan test for every SQL statement; a statement without one fails the build
Resource census	None	Thread, file-handle, child-process and memory counts taken before and after each suite; any growth fails
Panic hook	None	Route fuzzing and every long run install a hook that turns any panic into a failure with its input
Scale probe	`scale-probe.mjs` (new at HEAD): 1,000 turns, 100,000 messages, 2,000 sessions; not run by CI	Nightly lane with stored curves; a quadratic shape fails
Fixture repositories	`fixtures/repositories/parser-demo`	Polyglot monorepo, 1M-line legacy repo, multi-repo set, broken builds, embedded C firmware (STM32 HAL, ESP-IDF, Zephyr)

Harness piece	Today	Add
External services	In-process fakes for GitHub, providers and vendors	The same contract test runs against the fake and the real service, proving each fake still matches reality
IDE drivers	VS Code Extension Host E2E; JetBrains smoke programs and an EDT harness (new at HEAD) against a fake daemon	JetBrains real-IDE UI tests (IntelliJ Starter and Driver) on the two newest IDE versions, under the real look-and-feel
Agent eval harness	None: offline certification never uses a model	Task corpus with hidden tests, run on real models (see Agent quality evals)
Hardware simulation	None	Renode or QEMU boards with simulated sensors, plus a hardware-in-the-loop rig for releases

Core runtime suites

The runtime is the best-tested part of Faktor; the additions here are model-based, crash-at-every-point and property tests that search for cases nobody wrote down.

Feature	Must hold	Today	Add
Session state machine and journal	Replaying the journal rebuilds identical state; one `TurnCompleted` per genuine end; corruption is loud	Journal and recovery fault campaigns; `INV-JOURNAL-INTEGRITY`, `INV-DURABLE-CORRUPTION-LOUD`	Random command sequences against a reference model; kill at every failpoint, replay, compare state digests

Feature	Must hold	Today	Add
Typed ledger	Every entry decodes; head fold is replay-equivalent	Incremental open-time verification (new at HEAD)	Corrupt a row inside the verified prefix: a scheduled full sweep must fail loudly; 1,000,000-entry ledger opens in bounded time
Admission and idempotency	One run per submission id; a replay returns the same receipt	`INV-TASK-START-IDEMPOTENT`, `INV-PROMPT-START-IDEMPOTENT`	Same submission from two IDEs at once; crash between receipt and spawn; replay after restart
SQLite store	One writer, bounded readers, transactional migrations, restorable backups	313 store tests; plan tests	Every version migrates to the fresh head schema; restore each backup and compare; busy, disk-full and corrupt-page injection; concurrency check of the writer queue
Memory facts	Point lookups and kind-filtered pages stay O(page) under churn	Churn regressions at HEAD	Plan test on every fact query; 1,000,000 churn rows: activation, control queue and memory pages unchanged in result and time

Feature	Must hold	Today	Add
Checkpoints (CAS and snapshots)	Restoring gives back the exact tree; tampering is refused; cleanup never deletes a referenced blob	CAS fault campaign; `INV-CAS-COPY-TO-VERIFIED-STREAM`	Generated trees with symlinks, permissions, binaries, empty folders and 1 GB files round-trip; one flipped byte refused; cleanup during concurrent snapshots
Transactional editing	No patch lands on unexpected content; parse before accept; atomic writes	Edit fault campaign; 54 edit tests	A random concurrent writer always triggers the stale-hash refusal; parse-before-accept per indexed language; a crash mid-write leaves the old or new file, never a mix
Git worktrees and locks	One mutation lock per repository; leases are never stolen	66 git tests; `INV-GIT-LEASE-UNREADABLE-NOT-STOLEN`	Two daemons on one repo; stale lock after a crash; detached HEAD, submodules, sparse checkout, LFS pointers
Workspace files and watchers	Every path stays inside the workspace; watchers stay bounded	198 fs tests; root-cache bound (new at HEAD)	Symlink and junction escapes per OS; 1,000 shadowed turns with real watchers keep thread counts flat; the in-use durable root is never evicted

Feature	Must hold	Today	Add
Bounded context and compaction	Five memory classes stay separate; compaction meets its minimum reduction or is refused typed; the goal survives	`INV-COMPACTION-SUMMARY-CAP`; tiny-history refusal (new at HEAD)	10,000-turn soak with bounded growth; compaction never drops immutable instructions or task state; history sizes 0, 1 and 2 tokens all refused typed
Retrieval and index	Results fuse by rank; no match means no evidence; budgets hold	Production-wiring retrieval tests; tree-sitter for Rust, Python, TypeScript and JavaScript	Labelled queries per fixture repo with recall@10 floors; renaming a symbol moves its results; deterministic eviction under memory caps
Scheduler	No step starts before its dependencies; budgets never exceeded; bounded retries; breakers open and close	Scheduler DAG campaign; 71 tests	Properties over random DAGs; concurrency check of budget accounting; breaker state-machine model
Process supervision	No orphan processes on any OS, even if the daemon is killed	173 terminal, 107 PTY and 8 Job Object tests	Kill the daemon mid-build, then scan the process table per OS; double-forking and fork-bomb children stay contained
Shutdown	Zero live authorities after stop	`INV-SHUTDOWN-ZERO-LIVE-AUTHORITIES`	1,000 start/stop cycles with random work in flight; thread, handle and child counts return to baseline

Feature	Must hold	Today	Add
Memory, instructions, learning	Instruction roots are authoritative; learnings come only from verified episodes; skills load lazily	`INV-INSTRUCTION-ROOT-AUTHORITY`; 50 instruction and 43 learning tests	An unverified failure never becomes a learning; unused skills cost zero tokens; workspace memory never leaks across workspaces

Providers, tools and safety

The biggest additions here are a cross-provider differential test and corpora of hostile input, because most real failures come from the outside world.

Feature	Must hold	Today	Add
Model transports (OpenAI Chat and Responses, Anthropic, Gemini, Ollama, DeepSeek, gateways)	Every family yields the same normalized events for the same conversation; the agent has no provider-name branches	Recorded fixtures per family; static-authority scans	One scripted conversation encoded per family must normalize to equal event sequences; chaos-proxy cases; weekly live contract run per family on a small key budget

Feature	Must hold	Today	Add
Model catalog and probe	The catalog lists only what the endpoint serves; a probe never guesses capabilities	`POST /native/catalog/probe` (Revision 8) has no coverage row	Endpoint lies about a model, times out, returns 10,000 models, renames a model mid-session; Ollama `/api/tags` empty and huge
Model parameters	Compiled requests contain only parameters the model supports; unset and zero stay distinct	`model-parameters/compile` route; A03, A04	For every parameter × family: unset omitted, zero sent, unsupported refused before sending; the compiled request is byte-checked
Locality	Loopback, LAN and remote are reported truthfully; private-LAN inference obeys policy	A07, A08; control P-09 audited as missing	Endpoints on 127.0.0.1, ::1, 10/8, 192.168/16, a public IP and a DNS name that rebinds between checks

Feature	Must hold	Today	Add
Routing	Routes minimize cost to verified success; unknown quality authorizes nothing above floor 0	Router property crate; `INV-MAX-QUALITY-ONE-AUTHORITY`; 64-call history tail (new at HEAD)	Replay real outcome logs: never pick a model another beats on both cost and quality; tail and full history give the same decision
Vision and attachments	Non-vision models refuse before sending; media arrives byte-exact	Production-wiring vision test; A14	Malformed image headers, oversize files, per-family media encoding, an attachment deleted between admission and send
Embeddings	Semantic search is optional and degrades honestly	`no_embedder_degrades_honestly...` test	Embedder slow, wrong dimension, returning NaN, changing model mid-index

Feature	Must hold	Today	Add
MCP	Bounded queues, cancel exactly once, clean shutdown, trusted executables only	MCP invariants; A21, A22; remote transport and resources audited as missing	Misbehaving-server corpus (garbage frames, hangs, crash loops, oversized messages); OAuth expiry and revoke; a tool list that changes mid-turn
LSP and ACP	Workspace-scoped servers; bounded readers; exact framing	LSP and ACP invariants; official ACP client suite	Server crash during a request; 10 MB diagnostics; duplicate frame headers
Terminal and shell	Every change attributed to its command; crashes discovered; output bounded	`INV-SHELL-*`; 173 terminal tests	Commands that write outside the workspace, start background daemons, change directory, or print 1 GB

Feature	Must hold	Today	Add
Browser authority	First-party hosts only; downloads denied; screenshots on request only	8 `INV-BROWSER-*`; X-06, X-08, X-09 audited as missing	Hostile pages (redirect chains, third-party iframes, service workers, WebRTC); `validate-url` with IDN, IP literals and userinfo tricks
Sandbox and permissions	Capabilities never escalate; first run is audited	Sandbox and adversarial sandbox tests	Escape corpus per OS: symlink races, hard links, path traversal, `/proc`, environment injection; OS denial asserted on each platform lane
Network egress	One network authority; deny by default; secrets never leave	`INV-NETWORK-AUTHORITY-SINGULAR`; destination-policy fuzzing	Planted-secret corpus of 50+ key formats in files, environment and tool output, every one blocked; DNS rebinding; redirects to private addresses

Feature	Must hold	Today	Add
Prompt injection	Repo text, tool output, web or marketplace text and MCP replies are data, never authority	Security suite; sourcing security certification	Injection corpus across every input channel, asserting no permission, instruction or tool change; rerun weekly with real models
Hooks	A hook can allow, deny or annotate, never widen	22 hook tests; A24	Hook timeouts, crashes, malicious output, a hook that edits files; the hook test and delete routes (no test names them)

Engineering pipeline and proof

These suites protect Faktor's central promise: a task is finished only when its claims are independently proven. Each adds trap scenarios built to tempt a false "done".

Feature	Must hold	Today	Add
Task contract	Every significant task carries objective, criteria, constraints, invariants, risks, proof obligations and a change plan	Contract invariants (`INV-CONTRACT-*`)	Golden contracts for 30 fixture goals; rewording a goal must not drop a constraint (checked in the eval lane)

Feature	Must hold	Today	Add
Settlement identity	A turn settles only into the task it was admitted under	Bound on the main drive path (new at HEAD)	Interleaving test at every await point of the settlement tail; the queued, recovery and watcher paths that pass no task id; a static scan for lazy task-id reads
Task and run state	Every task state × session state pair maps to exactly one served tag	Tag unit tests (new at HEAD)	Exhaustive pair table, served by the daemon and rendered identically in both IDEs
System-intelligence registry	Tiers stay 0–5; findings are validated as data; a slow provider is dropped at its deadline; with no provider the native contract is unchanged	`INV-INTELLIGENCE-TIER-BOUNDS`; finding invariants; 102 semantic tests	Byte-identical contract with and without an absent provider; hostile findings (wrong snapshot, oversize text, invalid confidence)
Verification registry and Kiwi adapter	The native baseline is always present; a required external verifier that is missing is refused typed, never substituted	`INV-VERIFICATION-REQUIRE-PROVIDER`; `INV-KIWI-OPTIONAL-ADAPTER`	Forged verdicts (wrong identity, stale snapshot, oversized evidence) refused; Kiwi down mid-verification
Proof surfaces	Counts cover every durable row; missing, unavailable and corrupt stay distinct	Page-local count fix (new at HEAD)	Totals at 0, 1, 499, 500, 501 and 10,000 rows; a broken chain never renders verified

Feature	Must hold	Today	Add
Completion rules	Never complete because code compiles, tests pass, a file changed or an agent says so	Completion-review blocking; typed criterion proofs; A35	Traps: tests pass but a criterion is unproven; proof made on a stale tree; a reviewer that approves everything. Each must block completion
Completion steps (commit, push, PR)	Steps run in order and idempotently; a missing SCM is a durable blocker	Completion contract tests; fake GitHub	Crash between push and PR, then rerun: exactly one PR; branch-protection refusal; token expiry mid-step
Changes, review and restore	Accepting changes and restoring checkpoints never overwrite manual edits; previews match the result	Restore preview, change accept and review comments added in Revision 8 (no coverage rows); A33, A34	Preview then restore must produce exactly the previewed tree; a manual edit between preview and restore is refused; review comments survive restore and fork
Child agents	A child never exceeds its parent's budget or capabilities; controls apply exactly once	`INV-CHILD-*`; kind-filtered control reads (new at HEAD); A25, A26	Random delegation trees never exceed the root budget; steer, pause, resume, retry, stage and merge under concurrent calls; a parent crash leaves no running children

Feature	Must hold	Today	Add
Tournament, board, integration	Merging is an explicit approved step; owner edits invalidate it; the final tree is re-verified	Tournament and integration-record tests	Owner edit during landing; disagreeing candidates; verification runs on the merged tree, not the candidates
Budgets and settlement	Money is exact; a model call that would cross the cap never starts; settled spend never exceeds the cap	Accounting model check; settlement invariants; A30, A31	Cap of "1" on every entry path (plain prompt, task run, tournament, child); unknown pricing never reads as zero; crash between reservation and settlement; rounding at micro-unit edges
Engineering outcomes	One outcome record per run; replay never double-feeds the router	Outcome ledger tests; A36	Concurrent settlement and executor writing the same record; the VCSC denominator over cohorts with failed and unmeasured runs
Worker plane	One run per job generation; revocation is immediate; placement meets requirements	9 `INV-WORKER-*` invariants	Network partition between daemon and worker; lease-expiry races; a malicious worker returning forged results

Sessions, history, transcript and search

These surfaces changed most at HEAD, and the scale probe showed they break at real sizes. Every test here runs at the probe's sizes, not toy ones.

Feature	Must hold	Today	Add
Create, delete, export, fork, metadata	Delete purges with an audit tombstone; export holds everything and no secret; a fork shares no mutable state	Store deletion tests; delete and export routes added in Revision 8 have no coverage row; the metadata routes are named by no test	Delete during a running turn; export then re-import round-trips; every export scanned for planted secrets; fork then edit both sides
History listing	Every conversation reachable exactly once per walk	Cursor paging and index (new at HEAD)	Update conversations during a walk; 2,000 and 20,000 conversations; plan test for global and per-workspace listings
Message pages	Pages never shift, skip or repeat	100,000-message probe	Random appends during a page walk; `before` at 0, 1, the newest seq and past it
Transcript search	Finds prompts, answers and tool output; nothing from other sessions	Part rows included (new at HEAD)	Reference oracle over visible decoded text; JSON-key and quote queries; Unicode case folding; 100,000 rows within the page bound
Transcript rendering	Each tool call shows once with its true outcome	Denied-call merge (new at HEAD)	Window-edge orphan; every outcome × exit-code case; 16 sessions streaming at once with no cross-session text
Switching conversations	No write reaches the wrong session or host	A38; S72	Switch during streaming, during an approval prompt and during a compaction

Feature	Must hold	Today	Add
Turns, usage, efficiency	Per-session totals equal the sum of their provider calls; unknown cost stays unknown	Usage routes; K-12 audited as missing	Totals reconciled against `provider_call` rows for 1,000 turns; mixed known and unknown pricing

Configuration, credentials and settings

This area holds the riskiest untested routes: no test names the credential store, rotate, list or delete routes. Secrets get their own scanning rule: a planted secret must never appear in any response, log, export, event frame or crash dump.

Feature	Must hold	Today	Add
Config descriptors, search, provenance, snapshot	Every key has a descriptor; provenance names the winning layer	Config routes; 7 foundation config tests	Every key × machine, user, workspace and session layers: winner and provenance asserted; search finds every key
Preview, commit, reset, rollback, import, export, runtime acknowledgement	Changes are atomic (A37); a stale revision conflicts (S60)	Rollback, import and runtime acknowledgement are named by no test	Commit on a stale revision conflicts; crash mid-commit leaves old or new; hostile import (executable paths, secrets, unknown keys) refused; rollback to each earlier revision; acknowledgement only after restart-required keys apply

Feature	Must hold	Today	Add
Session settings	A per-session override applies from the next turn	Routes added in Revision 8 have no coverage row	Change during a turn applies to the next one only; override cleared on reset
Credential vault	Secrets are never returned, logged, exported or sent to a model; a deleted secret refuses reads	3 vault unit tests; store, rotate, list and delete routes named by no test	Planted-secret scan of every output channel; rotate during an in-flight call; delete then use gives a typed refusal; OS keychain unavailable
Provider connections	Test and health never report success they did not observe	Test, health and delete routes named by no test	Endpoint down, invalid TLS, wrong key, slow responses; deleting a connection a session is using
Service executable trust	A project cannot choose the executable; a checksum mismatch refuses start (A23)	O-04, O-05	Binary swapped between check and start; symlinked path; the updater replacing the binary
Workspace binding	A conversation binds to exactly one canonical workspace	S31	Two windows on one folder; renamed folder; network drive; case-insensitive file systems
Diagnostics	The export is redacted and bounded (O-03)	Preview and export routes	Planted secrets in every source are absent from the bundle; size cap holds on a 10 GB data folder

Feature	Must hold	Today	Add
Telemetry	Off by default; opt-in shows the payload (O-10)	Audited as missing	Once built: zero network calls while off; the preview equals what is sent

Foundation services

Instructions, memory, skills, hooks, MCP, presets, permissions and context run through `faktor-foundation`, the thinnest-tested crate. Each feature below needs a dedicated suite rather than the control exercise alone.

Feature	Must hold	Today	Add
Instructions	Precedence and conflicts are shown; pinned instructions survive compaction; the preview equals what the model receives	I controls audited partial	Precedence matrix; conflicting rules; hostile and 1 MB imports; byte-compare the preview with the sent request
Memory	Deletion is real (A16); retention runs; restore works within its window	Restore and retention routes named by no test	After delete, the record is absent from every store, index, export and model context; retention at the exact boundary; workspace scope never leaks
Skills	An untrusted skill never runs; installs are verified	6 of 7 skill routes named by no test; no cursor past 500	Hostile archives (path traversal, symlinks, huge files); invoking an untrusted skill refused; trust revoked when files change; 600 skills reachable

Feature	Must hold	Today	Add
MCP servers	Start, stop and connect are exact; auth expiry is recoverable (W08)	Start, stop, connect, tools and resources named by no test	Start twice; stop during a call; connect with an expired token; resources and prompts once built (T-12)
Presets and agents	Roles, fallback and envelopes are enforced	Roles and fallback routes added in Revision 8 have no coverage row	Fallback cycles; a deleted model in the chain; envelope caps enforced on children
Permissions and approvals	Approve once means once (A17); rules stay inside their boundary (A18); commands normalize (A19)	Rule delete named by no test	Rule-matching corpus (quoting, environment variables, `&&`, subshells, Unicode look-alikes); preview equals the real decision; a reply sent twice applies once; expired approvals
Context	Arithmetic is exact (A12); null is not zero (A11); sources keep identity (A13)	Compaction status and cancel named by no test	Pins survive compaction; manifest totals equal their parts; cancel mid-compaction leaves the original context

Enterprise, cloud, billing, workers and updater

These are disabled by default and power the paid plans, so they need the strictest money and identity tests. Ten `enterprise` and eight `updater` CLI subcommands are run by no test today.

Feature	Must hold	Today	Add
Organizations, members, invitations, SSO	Roles bound every route; an invitation works once	219 cloud tests; SSO tests	Role × route matrix; invitation reuse and expiry; OIDC state and nonce replay; clock skew; identity provider down
Enterprise settings, effective config, audit	Locked settings cannot be overridden; the audit log is append-only	Enterprise routes	Override attempts at every layer; tamper check over the audit chain
Retention, deletion jobs, tombstones	A deletion job advances exactly once; cleanup never deletes retained data	Deletion and GC routes	Crash mid-advance; GC racing a new retention hold; tombstone lists exactly what was purged
Billing, credits, entitlements, usage, outcomes	Money is exact; credits never go negative; overflow is checked	Billing tests; value-plan fixture	Concurrent credit grants; amounts at `i64::MAX`; usage fold equals the settlement ledger (details under Revenue product suites)
Workers and jobs	One binding per token; revocation is immediate; old generations are refused	9 `INV-WORKER-*` invariants	Token replay; heartbeat after revoke; result for a stale generation

Feature	Must hold	Today	Add
Updater	Only signed updates apply; active runs are preserved; rollback restores the old binary	165 updater tests; 8 CLI subcommands run by no test	Tampered package; power loss at each apply step; downgrade refused unless allowed; update while a turn runs
SCM webhook and GitHub App	Webhook claims are verified	SCM invariants	Forged signature; replayed delivery; installation id mismatch
Control-plane sign-in	Removing a local credential is distinct from revoking the remote session (O-07)	VS Code sign-in commands	Sign out while offline; token expiry mid-call

Sourcing engine

The offline sourcing suite is strong; what it lacks is proof against live vendors, which is the gate for selling it as the costing engine of the part-change product.

Feature	Must hold	Today	Add
Quote at quantity	Price honours breaks, minimum order, multiples, packaging, variants, promotions and account pricing	225 commerce tests, including plausible-wrong-price cases	Compare against a brute-force reference over generated offers; doubling the quantity never raises the unit price within one packaging when breaks are monotone

Feature	Must hold	Today	Add
Part matching	Exact part number wins; packaging suffixes stay distinct; ambiguity is reported, never guessed	`INV-COMMERCE-MPN-MATCH-EXACT`; `INV-COMMERCE-PART-IDENTITY`	Suffix corpus taken from real distributor catalogues; normalizing twice equals normalizing once
Money	No floating point anywhere; decimal strings on the wire	Float scan; money tests	Formatting then parsing returns the same amount for every currency and scale
Lazy exposure	Unrelated turns carry zero sourcing schema tokens, calls or browser launches	Release-blocking 10,000-turn token test	Near-miss phrases (a Rust `supplier` trait) and real chat logs added to the trigger corpus
Browser authority	Only first-party hosts; downloads denied; profiles confined; work stops at any human-verification page	8 `INV-BROWSER-*` invariants	Hostile pages: redirect chains, third-party iframes, service workers, WebRTC
Vendor changes	A changed vendor response is reported as drift, never misparsed	Offline fixture replay	Mutate every recorded fixture (renamed keys, moved fields, swapped types); each must yield drift or a correct parse
Live accuracy	Real vendors still parse and price correctly	Opt-in manual canaries	Weekly scheduled canaries on your own accounts, with recorded real basket totals as the expected answers

Sales gate: offer the sourcing engine to customers only after four consecutive weeks of green live canaries on Mouser, DigiKey and LCSC.

Protocol, CLI and IDE surfaces

The Edition 8 gates already turn the 200 controls, 72 screens, 16 workflows and 40 acceptance scenarios into checks. The gap is running them inside real IDEs and with real screen readers.

Surface	Must hold	Today	Add
Native protocol v1	Every route matches its frozen contract; strict bodies; money as decimal strings	Generated DTO parity; frozen contracts	The route template for all 246 routes (see the Routes tab); plugin versions N-1 and N-2 against the current daemon
Health and readiness	Health answers whenever the process runs; ready only after recovery	Health and ready routes	Ready stays 503 through a slow recovery; health never flaps during it
Event streams	Each frame delivered once per resume; per-session order	Stream probes; 16-stream perf workload (new at HEAD)	Reconnect at every event id; frames larger than the bound; a consumer that never reads
Legacy catalog routes	Only `GET /models` and `GET /capabilities` bypass the native path guard	Narrow client allowlist (new at HEAD)	Any other non-native path from the client is refused typed
CLI	Output, exit codes and help stay stable	398 CLI tests; doctor exit codes	The CLI template for all 43 commands (see the CLI tab)
VS Code webview lifecycle	Every reveal shows the complete served state	Handshake and backpressure fixes (new at HEAD)	Message before listener, hidden reveal, reload and rapid reveals; patches with an unknown type are counted, never silently dropped

Surface	Must hold	Today	Add
Controls and screens	All 200 controls and 72 screens behave per contract in both IDEs	Edition 8 registry, visual contracts, control exercise, optical, contrast and performance gates	Per-control contract tests derived from each record (see the Controls tab); run inside real IDEs, not only rendered webviews and offscreen Swing
Workflows and acceptance	W01–W16 complete and recover; A01–A40 pass in both IDEs	Edition 8 workflow and acceptance gates	Daemon crash and network loss mid-flow; one adversarial variant per scenario
Visual baselines	Every comparison uses a reviewed baseline	The pull-request lane writes a baseline when a platform has none	A missing reviewed baseline fails the lane; baseline changes ship rendered differences and a reviewer
Screen readers	Names, roles and announcements spoken correctly	Automated name and role checks; real screen readers recorded as not certified	Scripted NVDA, VoiceOver and Orca runs capturing spoken output per workflow
Install and update	Signed artifacts, verified installs, rollback	Install matrix; tamper test; 165 updater tests	Update while a run is active; rollback after a failed update; downgrade refused
Platforms	Each surface certified on each OS by its own run	JetBrains visual records missing on Windows	Windows JetBrains visual records; real-IDE lanes on macOS and Windows

Performance and scale

The scale probe added at HEAD found more real defects than any other suite in this commit, yet no CI lane runs it. It becomes a nightly gate with stored curves.

Workload	Size	Gate
Prompt turns, each followed by a forced compaction	1,000	Turn latency curve ratio at most 20 overall and 2 at steady state; p95 at most 15 s (the probe's own limits)
Transcript pages and search	100,000 messages	Every seq seen once; p95 page at most 500 ms; page curve ratio at most 6
History walk	2,000 conversations	Every conversation once, including during concurrent updates
Memory facts with edits, deletes and retention	5,000	Counts exact after each operation
Child agents in one run	500	One projection read, exact counts
Claims in one verification record	10,000	Totals exact; page bounded
MCP servers, skills, hooks	500 each	Every item reachable; listing bounded
Concurrent live event streams	16	No cross-session frames; bounded queues
Internal fact churn	1,000,000 rows (new)	Point lookups and kind pages flat; plan tests green
Shadowed turns with real file watchers	1,000 per OS (new)	Thread and handle counts flat

Every workload also records resident memory; growth beyond a stored per-platform baseline fails. Baselines are committed only after review, never written by the run that compares against them.

Agent quality evals

This is the largest gap. Offline certification proves the machinery works but never checks whether Faktor actually finishes real tasks, and it never runs a model.

Corpus. 200 tasks in the workloads Faktor targets:

- bug fixes
- features
- migrations and dependency upgrades
- security fixes
- multi-repo changes
- embedded part changes, for the hardware product

Each task has hidden acceptance tests, proof obligations and a reference patch. A fifth are written privately after the newest model's training cutoff, so no model has memorized them.

Runs. Each task runs three times on at least two model families, one hosted and one local through Ollama. Faktor's own spending limit caps the cost of each run.

Proving the combination. The same corpus also runs four ways: Faktor alone, with Tangerine, with Kiwi, and with both. The gap between those results is your evidence that the trio is worth more than its parts, and a regression alarm if it shrinks.

Metric	Definition	Gate
False completion rate	Faktor reports done but the hidden tests fail	1% or less; this is the product's core promise
Safety violations	Writes outside scope, secrets leaving, injected instructions followed	Zero
Verified resolve rate	Hidden tests pass and every proof obligation holds	No drop against the last release beyond run-to-run noise (one-sided test at 95%)
Honest disclosure	Unresolved risk is named whenever it exists	95% of applicable runs
Consistency	Share of tasks where all three runs succeed	Tracked per release

Metric	Definition	Gate
Cost per verified task	Spend divided by verified tasks	Alert when it rises 20%

Lanes. A 30-task smoke runs nightly, the full corpus weekly and on every release candidate. Results become a new signed `agent_eval` evidence gate beside `real_provider`. The 1%, 95% and 20% thresholds are my proposals.

Revenue product suites

Each revenue strategy gets its own suite, and each suite doubles as the sales proof for that product. Faktor already has a billing service and 12 value entitlements enforced at runtime boundaries, so the paid-plan suites start from real code.

Free core stays complete

- Run the whole community suite with billing disabled and no licence. Every non-entitlement test must give the same result as in a paid build.
- Build the Apache core with the paid crates removed; every community test must still pass.
- A dependency-direction scan fails if core code imports a paid crate. Today `deny.toml` requires every workspace member to be Apache-2.0, so that policy splits in two once paid crates move out.

Plans, entitlements and billing

Test	Oracle
Plan × feature matrix: 4 plans (local, pro, team, enterprise), an unknown plan and a foreign organization, against 12 value features (72 cells)	Each cell allows, or refuses typed at that feature's own runtime boundary, exactly as the plan table says
Entitlement change at each of the three admission points (new task, new child, new provider attempt)	Takes effect at the next admission point and never interrupts work in progress
Metering under crashes and retries	Usage events equal settlement rows exactly: none lost, none doubled

Test	Oracle
Invoice reconciliation	Usage ledger equals the invoice lines sent to the merchant of record, to the micro-unit
Licences	VS Code licence keys: signature and tamper checks. JetBrains Marketplace: trial, expiry, offline grace and seat counts

Kiwi for any agent

- **PR corpus:** pull requests written by other agents (Claude Code, Cursor, Copilot) and by humans, with planted defects. Kiwi must flag at least 90% of planted defects and raise false alarms on 5% or fewer of clean PRs (proposed thresholds).
- **Evidence:** a passing verdict without attached evidence is a test failure.
- **GitHub App:** webhook signatures verified, replays rejected, one verification per (repository, head commit).
- **Metering:** billed minutes match runner wall-clock time within 1%; infrastructure failures are never billed.
- **Conformance:** a published suite for the neutral `VerificationProvider` interface that any verifier can run.

Tangerine system intelligence

- **Call graph accuracy:** compare Tangerine's call graph for C and C++ with the ground truth from clang (`compile_commands.json`). Report precision and recall per relation family.
- **Honesty:** calls through function pointers must be reported as unknown, never as complete. A partly indexed tree must answer "partial".
- **Conformance:** a published suite for the neutral `SystemIntelligenceProvider` protocol that third parties can run.

Part-change product (hardware)

- **Corpus:** 25 real part changes on open firmware (Zephyr, ESP-IDF, STM32Cube):
  - sensor swaps with new register maps
  - flash chip changes
  - I2C address changes

    - pin-compatible MCU variants
    - passive parts that need no code change
- **Oracles:** a reference patch and hidden tests; cross-builds for every target; Renode simulation with sensor models; on release, two real development boards on a hardware-in-the-loop rig.
- **Traps:**
    - An incompatible replacement (voltage, pin-out, package) must be refused with the reason.
    - A deliberately altered datasheet fact must be caught.
    - A part change that needs no firmware change must produce no patch.
- **Cost:** the reported cost change must equal the sourcing engine's quote difference at the stated volume, and the report must validate against its schema.
- **Prerequisite:** the index's tree-sitter languages are Rust, Python, TypeScript and JavaScript today, so C and C++ support comes first.

Lanes, gates and release criteria

Five lanes split the work by cost and trust; a release needs every lane green at one commit.

Lane	Runs on	Suites	Blocks
Pull request	Every PR, Linux, no secrets	Format, lint, unit and property tests (fixed seeds), static scans, fault smoke, requirement-registry and surface-stub gate, query-plan tests, hygiene content sniffing, ratchets, short route fuzz under the panic hook, Edition 8 structural gates	Merge
Trusted push	`main`, on Linux, macOS and Windows	Full workspace, fault campaigns, the route and CLI templates for every surface, real-IDE UI tests, install matrix	Release candidates

Lane	Runs on	Suites	Blocks
Nightly	Schedule	Fault campaigns at scale, fuzzing, mutation sampling, concurrency and interleaving checks, scale probe, 1,000,000-row churn, resource census, soak, 30-task eval smoke	Next morning's merges until triaged
Weekly live	Schedule, on your own keys and accounts	Provider contracts, vendor canaries, GitHub App, Kiwi PR corpus, full eval, screen readers, Renode corpus	Sales of the affected product
Release	Tag	Everything above at one commit, plus hardware-in-the-loop and signed evidence	Shipping

A release ships only when all of these hold at the same commit:

- Every row in the requirement registry has a passing test, and no surface stub is still failing.
- No coverage row is delegation-only.
- The existing signed evidence gates pass (`cross_platform_lanes`, `real_provider`), plus five new ones: `agent_eval`, `live_contracts`, `assistive_tech`, `scale` and `hil`.
- No test is quarantined and no ratchet has grown.
- Every visual comparison used a reviewed baseline.
- Authority-bearing crates meet the mutation-score floor.
- The false completion rate is 1% or less and there are zero safety violations.

Build order

Close the HEAD findings first, then build the registry, because it turns every later gap into a visible red row. The evals come next, because they answer whether the product deserves money.

1. **HEAD fixes:** remove the core dumps from history, replace the `LIKE` filter, test the settlement paths that pass no task id, add the 1-micro cap regression, and schedule the scale probe.
2. **Requirement registry and surface stubs:** replace delegation-only rows, add the 17 missing rows, drop the 4 stale ones, and derive control status from code.
3. **Route template for all 246 routes**, starting with the 35 no test names; credential routes first.
4. **CLI template for all 43 commands**, starting with the 19 no test runs.
5. **Agent eval harness** with the first 50 tasks, including the four-way trio comparison.
6. **Property, state-model, interleaving and crash-at-every-point tests** for the journal, settlement, money, quotes and editing.
7. **Fake-versus-live contract lanes** for providers, GitHub and vendors; then the weekly sourcing canaries.
8. **Free-core and open-core boundary tests**, plus the plan × feature matrix and metering reconciliation.
9. **Kiwi PR corpus and GitHub App suite**, so Kiwi can be sold to users of any agent.
10. **C and C++ in the index and in Tangerine**, measured against clang.
11. **Part-change corpus, Renode simulation and the hardware rig.**
12. **Real-IDE JetBrains tests and scripted screen readers.**
13. **Mutation sampling across whole crates**, beyond the 158 invariants.

Sources

Read from the Faktor repository at its current `main`:

- HEAD commit ec50290 "Fixes"
- Scale probe
- Capability discovery
- Invariant coverage registry
- Edition 8 controls registry
- Native protocol v1
- Query-plan audit
- README: features, capability table, CI lanes
- Autonomous engineering specification

- Certification: commands, levels, evidence gates
- Architecture: billing, entitlements, worker plane
- Faktor Acquire specification
- Marketplace acquisition legal posture
- Invariant registry
- Edition 8 gates
- Value plan tiers fixture
- Tangerine system intelligence